Huawei Access Node YEL-ACC-67: Minor QoS / Congestion Event (ALARM_4352) Resolution SOP

Node: YEL-ACC-67 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_4352

Description

This failure domain occurs on the Huawei Access Node YEL-ACC-67 due to the occurrence of microbursts, interface tail drops, or QoS queue starvation, leading to a minor QoS congestion event detected by the network management system. This scenario is a result of the interaction between the node's hardware and software components, specifically the combination of the Enhanced Multi-Protocol Label Switching (e-MPLS) and the Quality of Service (QoS) mechanisms implemented on the device. In this context, the e-MPLS protocol facilitates label switching and forwarding decisions, whereas the QoS mechanism prioritizes and schedules traffic based on user-defined policies. However, when the node experiences a sudden surge in traffic, the QoS queue may become overwhelmed, leading to congestion and eventual tail drops. This failure mode is exacerbated by the specific hardware configuration of the Huawei Access Node YEL-ACC-67, which may not be adequately equipped to handle the increased traffic load, thereby perpetuating the congestion event.

Pre-requisite Checks

- Pre-requisite: Verify that the redundant link is operational and capable of carrying traffic in the event of a failure on the primary link., Status: Complete

- Pre-requisite: Gather all relevant logs and configuration files from the Huawei Access Node YEL-ACC-67 for further analysis., Status: Complete

Potential Root Causes

Cause	Probability
Hardware oversubscription or inadequate buffer size	High
Misconfigured QoS policies or incorrect traffic prioritization	Medium
Software bugs or firmware issues affecting the e-MPLS or QoS mechanisms	Low

Diagnostic Commands & Expected Outputs

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Command: display current-configuration, Expected Output: The current configuration of the device, including QoS and e-MPLS settings., Description: This command will provide an overview of the current configuration, allowing the analyst to identify any potential misconfigurations or settings that may be contributing to the congestion event.

Command: display interface status, Expected Output: The operational status of all interfaces on the device, including any errors or warnings., Description: This command will help identify any issues with the interfaces, such as high error rates or excessive packet
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drops, which may be related to the congestion event.

Command: display qos queue statistics, Expected Output: Detailed statistics on QoS queue performance, including packet drop rates and queue lengths., Description: This command will provide insight into the QoS queue performance, helping the analyst to determine if the queue is being overwhelmed and contributing to the congestion event.
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Resolution Steps

Step 1: Step: Verify and adjust QoS policies, Command: undo qos policy {policy_name}, Description: Revert any recently applied QoS policies to their default or previous configurations, and then reapply the policies with adjustments made to prioritize critical traffic and reduce congestion.

Step 2: Step: Increase buffer size or reduce oversubscription, Command: system-view\nbuffer {interface_name} {buffer_size}, Description: Increase the buffer size on the affected interface to accommodate more packets and reduce the likelihood of tail drops. Alternatively, reduce the oversubscription ratio to prevent excessive traffic from being forwarded to the interface.

Step 3: Step: Update firmware or software, Command: system-view\nsoftware {version_number}, Description: Update the firmware or software on the Huawei Access Node YEL-ACC-67 to the latest version, addressing any known bugs or issues affecting the e-MPLS or QoS mechanisms.

Post-Check Verification

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Command: display interface status, Expected Output: The interface status should be reset to normal, with no errors or warnings., Description: Verify that the interface status has returned to normal, indicating that the congestion event has been resolved.

Command: display qos queue statistics, Expected Output: The QoS queue statistics should show improved performance, with reduced packet drop rates and queue lengths., Description: Verify that the QoS queue performance has improved, indicating that the congestion event has been resolved.
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Roll-back Procedures

- Step: Revert QoS policy changes, Command: undo qos policy {policy_name}, Description: Revert any QoS policy changes made during the resolution process to their previous configurations.

- Step: Revert buffer size or oversubscription changes, Command: undo buffer {interface_name} {buffer_size}, Description: Revert any changes made to the buffer size or oversubscription ratio to their previous configurations.